

Deep Convolutional Neural Networks for Heart Sound Segmentation

Francesco Renna¹, Member, IEEE, Jorge Oliveira, and Miguel T. Coimbra², Member, IEEE

Abstract—This paper studies the use of deep convolutional neural networks to segment heart sounds into their main components. The proposed methods are based on the adoption of a deep convolutional neural network architecture, which is inspired by similar approaches used for image segmentation. Different temporal modeling schemes are applied to the output of the proposed neural network, which induce the output state sequence to be consistent with the natural sequence of states within a heart sound signal (S1, systole, S2, diastole). In particular, convolutional neural networks are used in conjunction with underlying hidden Markov models and hidden semi-Markov models to infer emission distributions. The proposed approaches are tested on heart sound signals from the publicly available PhysioNet dataset, and they are shown to outperform current state-of-the-art segmentation methods by achieving an average sensitivity of 93.9% and an average positive predictive value of 94% in detecting S1 and S2 sounds.

Index Terms—Phonocardiogram, segmentation, convolutional neural networks, hidden Markov models.

I. INTRODUCTION

CARDIAC auscultation is arguably the simplest, quickest, and most cost-effective first line of screening for a large number of heart conditions, including arrhythmia, valve disease, pulmonary hypertension, heart failure, etc. On the other hand, heart sounds are difficult to identify and analyze by a human listener, as they are faint, significant events are closely spaced in time, and their frequency content is at the lower end of the audible frequency range [1]. Extensive training is required to properly master cardiac auscultation and it was observed that, on average, only 20% of medical interns can effectively detect heart conditions using auscultation [2].

Manuscript received August 2, 2018; revised November 16, 2018; accepted January 15, 2019. Date of publication January 21, 2019; date of current version November 6, 2019. This work is a result and funded by the project NanoSTIMA (NORTE-01-0145-FEDER-000016), supported by Norte Portugal Regional Operational Programme (NORTE 2020), under the PORTUGAL 2020 Partnership Agreement, through the European Regional Development Fund (ERDF). It is also a result of the project DigiScope2 (POCI-01-0145-FEDER-029200), funded by Programa Operacional Competitividade e Internacionalização (POCI), in its FEDER component, and Fundação para a Ciência e Tecnologia, I.P., through national funds. It is also a result of the internal project SmartHeart in scope of project UID/EEA/50008/2013. This work was also funded by the FCT grant SFRH/BPD/118714/2016. (Corresponding author: Francesco Renna.)

The authors are with the Instituto de Telecomunicações, Faculdade de Ciências da Universidade do Porto, Porto 4169-007, Portugal (e-mail: frarena@dcc.fc.up.pt; oliveira_jorge@dcc.fc.up.pt; mcoimbra@dcc.fc.up.pt).

Digital Object Identifier 10.1109/JBHI.2019.2894222

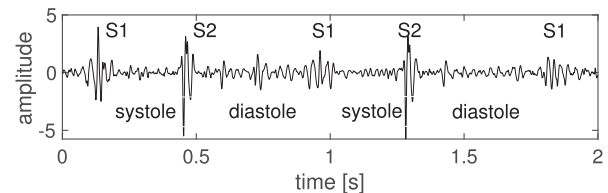


Fig. 1. Example of a recorded PCG signal in which the following four main heart sound components are shown: S1, systole, S2, and diastole.

The diagnostic potential and the simplicity of cardiac auscultation, combined with the scarcity of physicians able to extract useful and reliable diagnostic information when listening to heart sounds, have motivated the flourishing of recent research efforts in automatizing part or the entire process of analysis of the phonocardiogram (PCG) signal.

A key step required in the analysis of PCG signals is represented by the segmentation of heart sounds into their fundamental components. In each heart cycle it is possible to observe the first heart sound (S1), which is caused by the variation in blood pressure produced by the closure of the mitral and tricuspid valves, and their vibrations. The systolic interval follows immediately after the first heart sound and it corresponds to ventricular contraction. The systolic interval finishes at the beginning of the second heart sound (S2), which is generated by the closure of the aortic and pulmonary valves, and their vibrations. Finally, the second heart sound is followed by diastole (Fig. 1 reports a PCG example where the four main heart sound components are highlighted). The identification of these four fundamental segments in each heart cycle plays a key role in PCG analysis for different reasons: i) it allows the extraction of features targeted to each individual component for heart sound classification; ii) it allows the detection of the presence of extra sound components, e.g., the third and fourth heart sounds (S3 and S4), murmurs, ejection clicks, etc.; iii) it allows the extraction of useful information from the analysis of the morphology of the waveforms associated to the S1 and S2 sounds.

In contrast with the variety of approaches presented in the literature, that will be described in Section II, PCG segmentation still represents a challenging problem to solve when considering its application in real-world, noisy environments. In fact, the results of the 2016 PhysioNet Challenge on the classification of normal/abnormal heart sound recordings have shown that only a limited increase in classification performance can be achieved by using more sophisticated sound classifiers. On the other hand, improved segmentation algorithms are expected to be the best

point of entry to obtain more significant improvements in heart sound classification [3].

This work addresses this issue by proposing a novel heart sound segmentation approach, which is based on the use of a deep convolutional neural network (CNN). By leveraging a CNN architecture, the proposed method, unlike other deep learning solutions presented in the literature for heart sound segmentation, does not require a heart sound activity detection step to initially detect PCG segments before performing recognition, neither it relies on the extraction of *ad hoc* features from the signal. It can be applied directly to the PCG signal itself or to envelopgrams extracted from it. In this way, the sound features that minimize segmentation errors are learnt directly from training data by the CNN. In fact, the proposed approach, in contrast with other solutions presented in the literature, is not restricted to perform binary classification of S1 and S2 sounds, but it performs the entire heart sound segmentation task, thus automatically providing the location and the boundaries of fundamental heart sounds.

In addition, the choice of CNNs as a fundamental building block of the proposed method for heart sound segmentation is motivated by the recent success achieved by CNN architectures in related tasks, e.g., image segmentation [4]. In particular, a CNN-based PCG segmentation algorithm is expected to efficiently capture the characteristics of heart sounds corresponding to different PCG states, due to the ability of CNNs in modeling complex signal behaviors and their robustness to high signal variability.

On the other hand, CNNs do not incorporate *per se* prior information about the natural sequence of events occurring in heart sounds (e.g., an S1 sound is followed by a systole interval, which is followed by an S2 sounds, etc.). In fact, current state-of-the-art solutions for heart sound segmentation leverage the sequential nature of heart sound events and the corresponding temporal modulation by using statistical models such as the hidden Markov model (HMM) and hidden semi-Markov model (HSMM). These models are based on a sequence of hidden states generated via a Markov (or semi-Markov) process. Then, signal emissions are drawn according to probability distributions that depend on the particular state generated by the underlying Markov (or semi-Markov) process. In order to leverage the advantages provided by deep CNNs as well as sequential modeling methods as HMMs and HSMMs, these two approaches are combined in the segmentation algorithms proposed in this work.

In particular, this work proposes a novel segmentation approach that involves the following contributions:

- (1) The use of a 1-dimensional CNN architecture which is applied to different portions of the PCG signal. To the authors knowledge, this represents the first work that uses CNNs as a building block for PCG segmentation.
- (2) The development of temporal modeling solutions that combine the CNN outputs corresponding to the different portions of the PCG in order to produce the estimated segmentation sequence. Most notably, some of the proposed post-processing methods use CNN outputs to derive emission probabilities of a HMM and a HSMM.

- (3) A performance comparison between the proposed methods and other segmentation algorithms in the literature, including the current state-of-the-art, over the largest publicly available heart sound dataset with publicly available segmentation annotations.

The remainder of this paper is organized as follows: related works are described in Section II; the proposed segmentation methods are presented in Section III and the experimental methodology and results are reported in Section IV. Finally, conclusions are drawn in Section V.

II. RELATED WORK

Several solutions have been proposed in the literature to perform PCG segmentation (see [5] for a more complete analysis). These can be roughly divided into three classes of algorithms: peak-picking algorithms, methods based on the extraction of features and the use of classifiers, and methods that use sequential models.

The first class of segmentation algorithms is based on the extraction of envelopgrams from PCG signals and the use of peak-picking algorithms to estimate the position of the principal heart sounds S1 and S2. Such envelopgrams can be defined over different domains. Some are extracted in the time domain, as the Shannon energy introduced in [6], or the squared energy introduced in [7]. Envelopgrams can be extracted in the frequency domain, as done by the method described in [8], which computes the Shannon energy of the local spectrum, which is evaluated via the S-transform. Other domains considered in the literature include the Hilbert transform [9] and different wavelet transforms [10], [11]. The approaches belonging to this first class of segmentation algorithms attempt to transform the recorded heart sound signals into domains where waveforms associated to fundamental heart sounds are emphasized, whereas noise components are reduced. However, the implementation of peak-picking algorithms involves the use of *ad hoc* peak-rejection heuristics to combat the effect of in-band noise. Also, such heuristics are not always guaranteed to be robust when applied to heart sounds collected in different conditions and with different devices.

The second class of algorithms involves the extraction of *ad hoc* features from different portions of the PCG signal and the use of classifiers to determine if such features belong to fundamental heart sounds (S1, S2) or not. Different kinds of features are used in this sense: time domain features [12], frequency domain features [13], wavelet features [14], etc. Then, various types of classifiers have been adopted in the literature. From simple k-means clustering [15], [16] to decision trees [17], from support vector machine (SVM) classifiers [14], to artificial neural network (ANN) classifiers [18], [19]. More recently, [20] proposed the use of fully connected deep neural networks to distinguish S1 sounds from S2 sounds.

All these algorithms, however, first require a heart sound activity detection step to identify PCG segments that are possible candidates to be labeled as S1 or S2 sounds. For example, for the segmentation algorithm described in [20], S1 and S2 candidates are first identified using the segmentation algorithm proposed in [6]. Moreover, the performance of these algorithms strongly

depends on the choice of the *ad hoc* features extracted from the different PCG segments.

The third class of segmentation algorithms leverages statistical models to include prior information about the sequential nature of PCG signals. Recently, stochastic sequential neural networks have been proposed [21]. However, such approach is not specifically tailored for heart sound segmentation and is shown to achieve results that are significantly below the current state-of-the-art (average error rate of 29.29% in predicting the latent variables associated to the different segments of the heart sounds). On the other hand, state-of-the-art segmentation performance has been achieved by using HMMs and HSMMs. For both models, the sequence of segments composing a heart sound recording is mapped to a sequence of hidden states, which is generated via a Markov or semi-Markov process. Then, the recorded signal observations are modeled via statistical distributions that depend over time on the particular state generated by the underlying Markov (or semi-Markov) process. In particular, HMMs have been adopted for heart sound segmentation by [22], and then by [23], which proposed to use an HMM to classify peaks picked from the homomorphic envelopogram of the PCG. Later, [24] extended this approach by using Gaussian mixture model (GMM) priors to model the emission distributions.

When using discrete-time HMMs, the statistical distributions of the time periods spent in the different states of the Markov model, i.e., the sojourn times, are implicitly modeled by geometric distributions with parameters depending on the state transition probabilities. On the other hand, HSMMs represent a generalization of HMMs, where the sojourn time distributions are not forced to be geometric. Schmidt *et al.* have introduced the use of HSMMs for PCG segmentation [25], [26], thus guaranteeing a relevant performance improvement with respect to HMM-based solutions. Then, other works have improved further the performance of HSMM-based segmentation algorithms by proposing methods to adapt sojourn time distributions to the different PCG signals [27], [28] or by refining the models adopted for emission distributions. Namely, in [29], SVM-based emission distributions were proposed and later [30] adopted emission distributions based on the logistic regression function. This latest solution, jointly with a modified Viterbi algorithm that addresses boundary conditions, made [30] the current state-of-the-art algorithm for heart sound segmentation. Such algorithm has also been adopted as the reference segmentation method for the 2016 PhysioNet Challenge on the classification of normal/abnormal heart sounds [5], [31].

Then, this work aims to propose a method that leverages the flexibility provided by CNNs in modeling complex data with high variability in conjunction with the use of statistical sequential models (namely, HMMs and HSMMs) which have been shown to describe very effectively the temporal evolution of PCG signals. In this way, a segmentation performance superior to the current state-of-the-art can be achieved, thus paving the way to an improved potential in extracting useful information from heart sound recordings.

Before describing in details the proposed CNN-based method for heart sound segmentation, it is relevant to note in passing that, recently, deep neural networks and deep convolutional neural

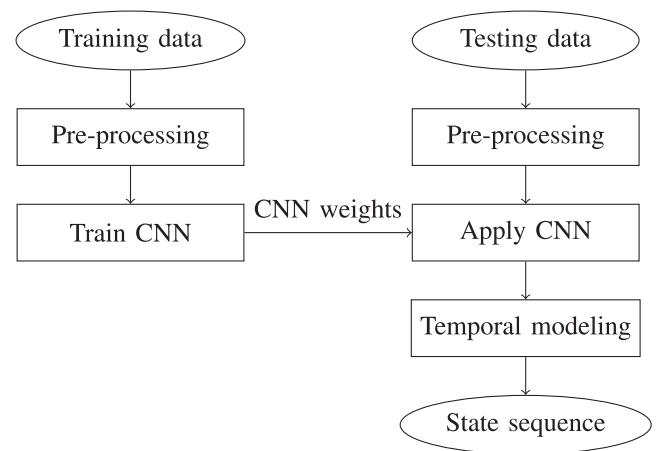


Fig. 2. Diagram of the proposed segmentation method: training phase and testing phase. The training phase is used to determine the weights of the CNN that will be applied to heart sound signals in the testing phase.

networks have been used to perform heart sound classification (see, for example, [32]–[39]). However, these algorithms are designed to discriminate between normal and abnormal heart sounds, thus they do not provide any information about the position of S1 and S2 sounds within the recorded PCGs. On the other hand, to the authors’ knowledge, the methods described in this work represent the first use of deep convolutional neural networks specifically designed for the signal processing task of heart sound segmentation.

III. METHODOLOGY

In this section, the three main steps of the proposed segmentation algorithms are described in detail. Each proposed approach consists of a training phase and a testing phase. Signals involved in both training and testing are pre-processed according to the methods described in Section III-A. Labeled training data is used to determine the parameters (weights) that define the operations implemented by the CNN. In the testing phase, the trained CNN is applied to the pre-processed testing data, and the corresponding output undergoes a further post-processing stage in order to generate the estimated state sequence associated to each heart sound in the testing set (see Fig. 2). In particular, four different temporal modeling strategies will be presented: sequential max temporal modeling, HMM temporal modeling, HSMM temporal modeling, and adaptive HSMM temporal modeling. The advantages and disadvantages of the considered temporal modeling strategies will be discussed in Sections IV-C and IV-D.

A. Pre-Processing

PCG signals are first filtered with high-pass and low-pass Butterworth filters of order two with cut-off frequencies equal to 25 Hz and 400 Hz, respectively. Then, the spike removal procedure described in [26] is applied to the filtered signals. The four envelopograms/envelopes considered in [30] are extracted from the filtered signals: i) homomorphic envelopogram, ii) Hilbert envelope, iii) wavelet envelope, and iv) power spectral density (PSD) envelope. Such envelopograms are then

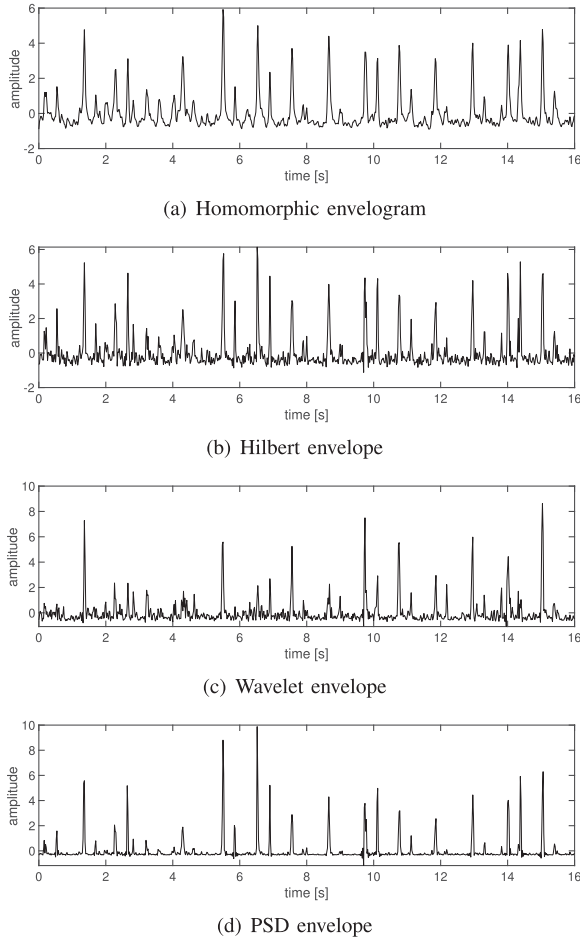


Fig. 3. Example of the envelopes/envelopes extracted from the PCG signals.

downsampled to 50 Hz [30]. Finally, all the four envelopes/envelopes are normalized in order to have zero mean and unit variance. As shown by the example reported in Fig. 3, the four envelopes/envelopes considered in this work provide complementary information about the PCG signal from which they are extracted, as they represent different tradeoff levels between noise rejection and amplitude resolution.

For each heart sound, the normalized envelopes are collected in the 4-dimensional signal $\mathbf{x}(t)$, where $\mathbf{x}(t) \in \mathbb{R}^4$ for $t = 0, \dots, T-1$, and where t indicates the time instant. Then, $s(t)$ is defined as the sequence containing the state labels associated to each time instant, i.e., $s(t) \in \{0, 1, 2, 3\}$, where state 0 corresponds to an S1 sound, state 1 to a systole interval, state 2 to an S2 sound, and state 3 to a diastole interval.¹ Then, given a heart sound signal $\mathbf{x}(t)$, the objective of the segmentation algorithm is to provide an estimate of the corresponding state sequence $s(t)$.

Four-dimensional patches of fixed length N are extracted from the signal $\mathbf{x}(t)$, with a given stride τ . Such portions of the signal $\mathbf{x}(t)$ represent the inputs of the CNN. They are denoted

¹In this work, we consider only the 4 main PCG components, i.e., S1, systole, S2, and diastole.

by $\mathbf{X}(n) \in \mathbb{R}^{N \times 4}$ and they are obtained as

$$\mathbf{X}(n) = \begin{bmatrix} \mathbf{x}(n \cdot \tau) \\ \vdots \\ \mathbf{x}(n \cdot \tau + N - 1) \end{bmatrix}, \quad (1)$$

for $n = 0, \dots, \lfloor \frac{T-1-N}{\tau} \rfloor$, where $\lfloor x \rfloor$ denotes the greatest integer lower than or equal to x .

B. Convolutional Neural Network Architecture

Various CNN architectures have been presented in the quickly growing deep learning literature. This work proposes the use of a CNN architecture which is inspired by the U-net originally presented in [4] for image segmentation. The proposed architecture, in contrast with the original U-net, is applied to 1-dimensional signals with four different channels, where each channel corresponds to one of the four envelopes extracted from the PCG. Moreover, the number of filters contained in each convolutional layer has been adapted to better capture the inherent structure of the input signals, i.e., patches extracted from PCG envelopes. The proposed architecture is reported in Fig. 4.

Such deep network contains convolutional layers which are followed by rectified linear unit (ReLU) activation functions [40]. The convolutional layers implement 1-dimensional convolutions of their inputs with different sets of filters (feature maps). The filter corresponding to each feature map of a convolutional layer is defined by 3 weights. More specifically, on denoting by N_ℓ the spatial dimension of the input of the ℓ -th convolutional layer and by k_ℓ the number of output feature maps of the ℓ -th convolutional layer, so that the input and output of such layer can be collected in the matrices $\mathbf{W}^\ell \in \mathbb{R}^{N_\ell \times k_{\ell-1}}$ and $\mathbf{Z}^\ell \in \mathbb{R}^{N_\ell \times k_\ell}$, it holds:

$$\mathbf{Z}_{i,j}^\ell = \sum_{i'=1}^3 \sum_{j'=1}^{k_{\ell-1}} \mathbf{A}_{i',j'}^\ell \mathbf{W}_{i+i',j'}^\ell, \quad (2)$$

where the notation $\mathbf{A}_{i,j}$ is used to denote the element in the i -th row, j -th column of the matrix $\mathbf{A}$. The inputs of convolutional layers are zero-padded so that the corresponding outputs have the same spatial dimensions as the input. Note also that the elements of the matrices $\mathbf{A}^\ell$ are the weights corresponding to the different filters that produce the different feature maps in the ℓ -th convolutional layer. Therefore, on denoting by L the total number of convolutional layers in the network, the set $\theta = \{\mathbf{A}^1, \dots, \mathbf{A}^L\}$ contains all the parameters that are learnt during the training phase. Then, the output of all convolutional layers, except the last layer, are followed by a ReLU activation function, which operates entry-wise and returns that value $\max\{\mathbf{Z}_{i,j}^\ell, 0\}$ for all i, j .

Moreover, the proposed architecture includes max pooling layers, which are responsible for downsampling (by a factor 2) the outputs of middle layers, as well as upsampling layers (by a factor 2) which are interleaved with the late convolutional layers. This choice is fundamentally motivated by the fact that the adopted CNN structure implements a dyadic scale decomposition, which has a nice analog with a multiresolution wavelet decomposition, which has been shown to be effective

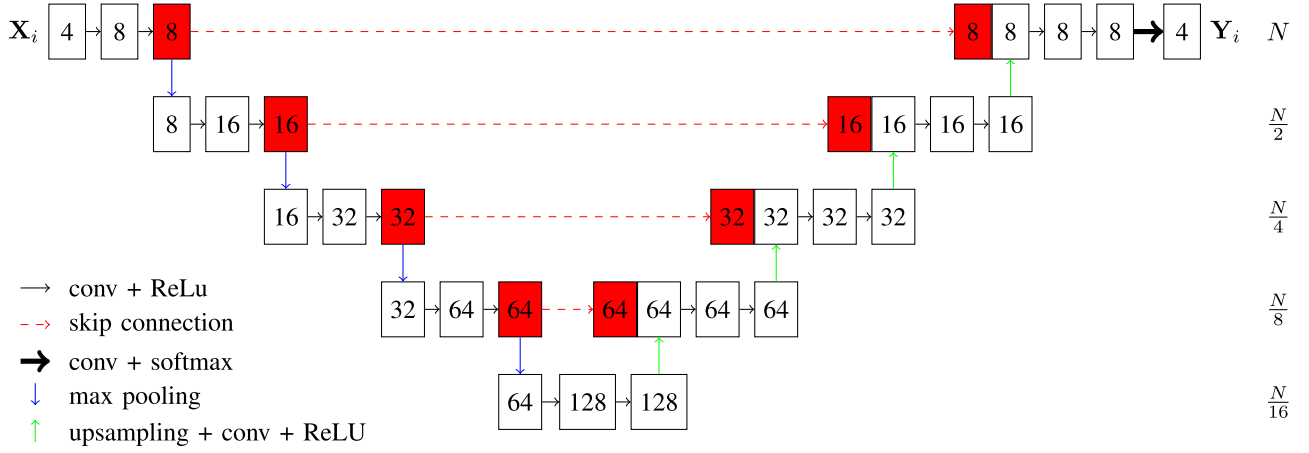


Fig. 4. Architecture of the deep CNN used in the proposed segmentation algorithms. The numbers inside the boxes in the diagram represent the number of feature maps in the corresponding convolutional layer. The numbers on the right hand side of the figure indicate the spatial dimension of the inputs and outputs of the layers contained in the corresponding row.

for PCG segmentation [10], [11]. In fact, such CNN implements an encoder-decoder architecture, in the sense that the outputs of the middle layers offer a compact representation of the input signals in a low-dimensional manifold which contains the main information about the segmentation state of the PCG, thus reducing the impact of noise and signal variability. Moreover, it is possible to note that the overall receptive field of the proposed CNN is large, which means that information coming from neighboring portions of the input signal is effectively combined when deciding the state of a particular sample drawn from the PCG signal under test.

Skip connections are also inserted in the network, in order to allow direct transfer of information from the first layers to the late layers. Such connections are implemented by simply copying the information contained in the feature maps of the first layers and including it as additional feature maps of the late layers.

Finally, the last convolutional layer implements 4 feature maps with a single weight each, and it is followed by a softmax activation function [40]. Therefore, on denoting by $\mathbf{Z}^L$ the output of the last convolutional layer, i.e., the input of the softmax activation, and, on denoting by $\Phi_\theta(\cdot)$ the function implemented by the proposed CNN, where θ represents the set of weights that define the convolutional layers in the network, the CNN output corresponding to the input $\mathbf{X}(n)$ is given by

$$\mathbf{Y}(n) = \Phi_\theta(\mathbf{X}(n)), \quad (3)$$

where

$$\mathbf{Y}_{i,j}(n) = \frac{\exp(-\mathbf{Z}_{i,j}^L)}{\sum_{j'} \exp(-\mathbf{Z}_{i,j'}^L)}. \quad (4)$$

Then, each row of the matrix $\mathbf{Y}(n) \in \mathbb{R}^{N \times 4}$ contains the probability that the corresponding sample of the PCG signal belongs to state 0, 1, 2, or 3.

An example of the output generated by the CNN for a PCG patch of approximately 2.4 seconds is reported in Fig. 5, where the different lines correspond to the probability that each sample in the patch belongs to a given state.

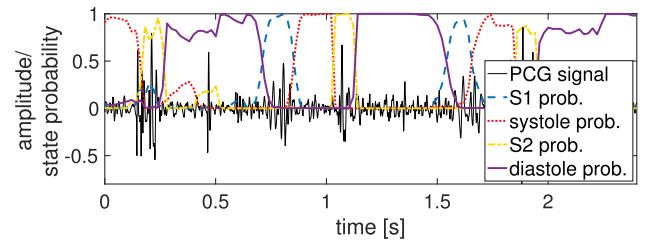


Fig. 5. CNN output example: PCG signal and state probabilities for each sample in the patch.

C. Temporal Modeling

Depending on the values of the patch size N and the stride τ , for a given time instant t of the heart sound signal there can be different output matrices $\mathbf{Y}(n)$ containing the probabilities that such time instant belongs to one of the 4 available states. In fact, overlapping patches are used in order to minimize the impact on segmentation of data samples near the border of each patch. Therefore, the information obtained from different overlapping patches is combined by simply averaging the state probabilities associated to different output matrices $\mathbf{Y}(n)$. The so obtained probabilities are collected in the matrix $\mathbf{B} \in \mathbb{R}^{T \times 4}$, which is such that $0 \leq \mathbf{B}_{t,j} \leq 1$ for $t = 0, \dots, T-1, j = 0, 1, 2, 3$ and $\sum_{j=0}^3 \mathbf{B}_{t,j} = 1$, for $t = 0, \dots, T-1$.

A first, coarse estimate of the state sequence $s(t)$, which is denoted by $\tilde{s}(t)$, can be obtained from the matrix $\mathbf{B}$ by choosing the state corresponding to the maximum probability among S1, systole, S2, or diastole, i.e., by defining $\tilde{s}(t)$ as

$$\tilde{s}(t) = \arg \max_j \mathbf{B}_{t,j}. \quad (5)$$

This operation represents the standard way CNN outputs are used for classification tasks. On the other hand, it is straightforward to note that such state sequence $\tilde{s}(t)$ does not incorporate any prior information about the fact that only few transitions are available between different states, as an S1 event is always followed by a systole event, a systole event is always followed by an S2 event, etc.

In this section, four different temporal modeling strategies will be presented, which force the output of the segmentation algorithm to be consistent with admissible transitions among heart sound states. Such four methods represent different trade-offs between computational complexity and the amount of prior information included in the modeling procedure about the temporal evolution of the PCG signal.

1) Sequential Max Temporal Modeling: The first method considered to force the output sequence $\hat{s}(t)$ to contain only admissible transitions among states is obtained by applying the following definition: $\hat{s}(0) = \tilde{s}(0)$, and

$$\hat{s}(t) = \begin{cases} \tilde{s}(t), & \text{if } \tilde{s}(t) = (\hat{s}(t-1) + 1) \bmod 4 \\ \hat{s}(t-1), & \text{otherwise,} \end{cases} \quad (6)$$

for $t > 0$. Note that the output state sequence $\hat{s}(t)$ now contains only admissible state transitions, thus providing a consistent way to localize the different segments of the heart sounds. The main advantage of this method is represented by its low computational complexity and the possibility of applying such algorithm in real time, as the segmentation output at a given time t does not require the knowledge of the CNN outputs corresponding to the entire PCG signal.²

2) HMM-Based Temporal Modeling: More sophisticated strategies can be used in order to force the output sequence of the proposed segmentation algorithm to be compatible with admissible state transitions. In fact, it is possible to use in a more refined way the posterior state probabilities obtained at the output of the CNN that are contained in the matrix $\mathbf{B}$.

In particular, it is possible to consider such output probabilities in conjunction with an underlying HMM that modulates state transitions. Such HMM is described by the following parameters: initial state probabilities π , state transition probabilities $\gamma_{i,j} = p(s(t) = j | s(t-1) = i)$, and emission probabilities $e_{t,j} = p(\mathbf{x}(t) | s(t) = j)$. In this case, estimates for the initial state probabilities π and the state transition probabilities $\gamma_{i,j}$ can be inferred from annotated training data by using standard maximum likelihood (ML) estimators [41].

On the other hand, the values of the emission probabilities corresponding to a particular heart sound under test are obtained from the corresponding $\mathbf{B}$ matrix as follows. On assuming that the values in such matrix represent a good approximation of the posterior state probability at time t given the observation of the emission vector $\mathbf{x}(t)$, i.e.,

$$p(s(t) = j | \mathbf{x}(t)) \approx \mathbf{B}_{t,j}, \quad (7)$$

on leveraging the Bayes theorem, it is possible to compute emission probabilities as

$$e_{t,j} = p(\mathbf{x}(t) | s(t) = j) = \frac{p(s(t) = j | \mathbf{x}(t)) \cdot p(\mathbf{x}(t))}{p(s(t) = j)}. \quad (8)$$

In this case, $p(\mathbf{x}(t))$ is approximated via a multivariate Gaussian distribution whose mean and covariance matrix are estimated from training data using standard ML estimators. Moreover, the

prior state probabilities $p(s(t) = j)$ are set to be equal to the initial state probabilities contained in π .

Then, the so defined HMM can be used to efficiently determine the state sequence that maximizes the likelihood

$$\mathcal{L}(s, \mathbf{x}) = p(s(0), \dots, s(T-1), \mathbf{x}(0), \dots, \mathbf{x}(T-1)), \quad (9)$$

via the Viterbi algorithm [41]. Therefore, the output of the proposed segmentation method $\hat{s}(t)$ is obtained as the sequence that maximizes the likelihood.

3) HSMM-Based Temporal Modeling: The emission probabilities computed according to the method described in Section III-C2 can also be used in conjunction with a more general HSMM. In this case, sojourn time distributions are assumed to be Gaussian, where the corresponding means and variances are inferred from the analysis of the autocorrelation function of the homomorphic envelopogram extracted from the PCG signal under test. The way such parameters are estimated was introduced in [26] and later adopted also in the state-of-the-art solution described in [30].

Then, the output sequence $\hat{s}(t)$ of the proposed HSMM-CNN segmentation methods is obtained as the sequence which maximizes the likelihood (9), which is efficiently computed using the Viterbi algorithm [42].

4) Adaptive HSMM-Based Temporal Modeling: Also this temporal modeling method is based on the use of HSMMs with emission probabilities computed according to the method described in Section III-C2. On the other hand, in this case, sojourn time distributions are still assumed to be Gaussian, but the corresponding means and variances are estimated from the PCG data using the adaptive procedure described in [28]. Such procedure returns the sojourn time distribution parameters that better fit the sequence of observations (PCG samples) by maximizing the incomplete likelihood associated to the sequence $\mathbf{x}(t), t = 0, \dots, T-1$ (see [28] for details). The solution of such maximization problem is approximated via the use of an iterative expectation maximization (EM) approach.

Then, after determining the most suitable set of sojourn time parameters, the output sequence $\hat{s}(t)$ of the adaptive HSMM-CNN segmentation method is chosen as the maximum-likelihood sequence, which is computed using the Viterbi algorithm.

IV. EXPERIMENTS

The proposed segmentation algorithms are compared with other segmentation algorithms appeared in the literature which are based on temporal modulation. Namely, the algorithm described in [26] and the HSMM-based method proposed in [30], which is currently considered as the state-of-the-art PCG segmentation algorithm.³

The performance of the proposed CNN-based segmentation algorithms and of the other segmentation methods considered is tested via 10-fold cross-validation on the available heart sound data. Namely, the available heart sound dataset is divided into

²A numerical comparison of the computational complexity of the different segmentation methods considered in this work is reported in Section IV-D.

³The HSMM-based segmentation algorithm in [30] was implemented using the code made available online by the authors at <https://physionet.org/physiotools/hss/>

10 subsets. Then, tests are performed by selecting iteratively 1 out of these 10 subsets as the testing sets, while using the remaining 9 subsets for training. Note that subsets are divided so as to guarantee that sounds from patients contained in the testing set are not contained in the training set, in order to avoid overfitting.

The CNN used in the proposed segmentation methods is trained with different patch dimensions ranging from $N = 64$ samples (approximately 1.2 seconds) to $N = 512$ samples (approximately 10.2 seconds). The patches are extracted from the training recordings with strides from $\tau = 8$ to $\tau = 128$ samples. In all the experiments, the weights of the CNN are learnt using the categorical cross-entropy loss function [40] and the Adam optimizer [43] with learning rate equal to 10^{-4} . At the beginning of the training procedure, for each fold in the 10-fold cross-validation setup, the network weights are initialized using Glorot's uniform initializer [44]. The mini-batch size is set to 1 and the following early stopping strategy is adopted, in order to avoid overfitting. For each of the 10 folds in the testing setup, the remaining 9 folds are used for training and validation: namely, 90% of that data is used for training and 10% for cross-validation. The number of training epochs is fixed to 15 and, at the end of each epoch, the cross-entropy loss function is evaluated over the validation data with the current network weights. Finally, the weights obtained after the epoch corresponding to the minimum loss-function value on the validation data are retained for testing. Note that the data used for validation is not used to train the network.

The CNN considered in this work was implemented using Keras [45] with TensorFlow [46] as a backend. Each model required approximately 3.5 hours for training, when using a workstation with an intel 6-core processor, 64 GB of RAM and mounting an Nvidia Titan Xp GPU.

A. Performance Metrics

A first performance metric used in this work is the sample accuracy (A), which represents the fraction of samples in the output sequence $\hat{s}(t)$ that are correctly allocated to the corresponding state in the ground truth sequence $s(t)$.

Two further metrics are used to evaluate the performance of the proposed algorithms in determining the position of the fundamental heart sounds S1 and S2: positive predictive value (P_+) and sensitivity (S). Such metrics are computed according to the description in [26], i.e., a true positive (T_p) is counted when the center of an S1 (S2) sound in the estimated sequence $\hat{s}(t)$ is closer than 60 ms from the center of the corresponding S1 (S2) sound in the ground truth sequence $s(t)$. All other S1 and S2 sounds in the estimated state sequence are considered as false positives (F_p). Then, the positive predictive value P_+ is given by:

$$P_+ = \frac{T_p}{T_p + F_p}. \quad (10)$$

On the other hand, the sensitivity S is defined as:

$$S = \frac{T_p}{T_{tot}}, \quad (11)$$

where T_{tot} represents the total number of S1 and S2 sounds in the ground truth state sequence $s(t)$.

All performance metrics are computed for each recording in the test set and then averaged over the test set. Finally, the values corresponding to the different 10 test subsets are reported in Section IV-C.

B. Materials

The heart sound segmentation algorithms considered in this work are tested over the largest publicly available heart sound dataset provided with segmentation annotations. This is the dataset that was made publicly available for the PhysioNet/CinC challenge 2016. The dataset contains 792 heart sounds recorded from 135 patients in different clinical and non-clinical environments.⁴ From those, 406 sounds are collected from patients with pathological heart lesions (most commonly mitral valve prolapse), as assessed by echocardiography. The remaining 386 sounds are collected from healthy patients. Sound recordings have variable durations in the range from 1 to 35.5 seconds and they are sampled at 1 kHz. They are collected from several spots over the chest and they are possibly corrupted by different sources and noise levels. The annotations provided with the dataset are computed via the analysis of synchronous ECG recordings, based on the agreement between five different automatic R-peak and end-T-wave detectors [5].

C. Results

Experimental results are first obtained by considering CNN approaches with $N = 64$ samples and $\tau = 8$ samples. In this case, 786 heart sound recordings with duration greater than 1.2 seconds are considered. In Figs. 6, 7, and 8 are reported the boxplots corresponding to the values of sample accuracy (A), sensitivity (S), and positive predictive value (P_+), respectively, achieved by the segmentation algorithms presented in [26] and [30], and by the proposed CNN-based methods.

Then, in order to investigate the performance of the proposed CNN-based algorithms over heart sound signals with different durations, further results were collected by considering separately the cases when $N = 128$, 256, and 512 samples, therefore considering heart sounds with duration greater than 2.5, 5.1, and 10.2 seconds, respectively. The number of heart sounds considered in the three cases is 590, 427, and 314, respectively. In all cases, the stride τ is chosen as $1/8$ of the patch dimension, thus corresponding to $\tau = 16$, 32, and 64 samples. Table I reports the mean and the standard deviation of the performance metric values corresponding to the different segmentation algorithms considered in this work, obtained from 10-fold cross-validation over the corresponding heart sound dataset.

It is possible to observe that the proposed CNN-based algorithms guarantee higher average accuracy, sensitivity, and positive predictive value than the temporal modulation segmentation algorithms in [26] and [30], for all the considered metrics and all the considered setups. Moreover, such performance difference is statistically significant in the majority of the considered

⁴The sounds are available online at <https://PhysioNet.org/physiotools/hss/>

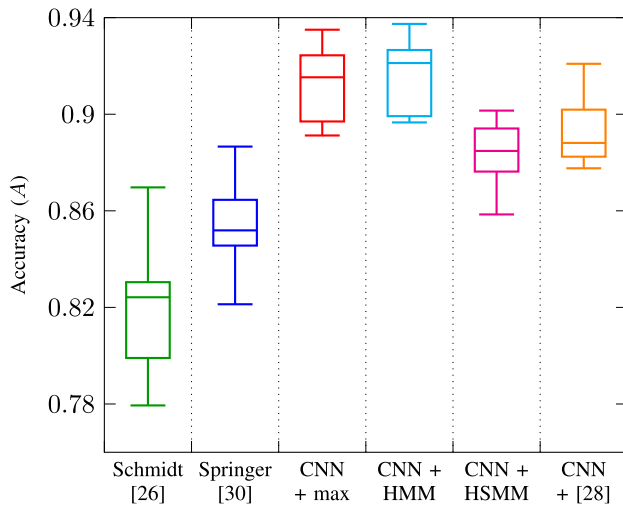


Fig. 6. Sample accuracy (A) of the segmentation algorithm by Schmidt *et al.* [26], the algorithm by Springer *et al.* [30] and the proposed CNN-based approaches: CNN approach with sequential max temporal modeling (CNN + max), CNN approach combined with an HMM (CNN + HMM), CNN approach combined with an HSMM with sojourn time parameter estimation as in [30] (CNN + HSMM), CNN approach combined with an HSMM with sojourn time parameter estimation as in [28] (CNN + [28]).

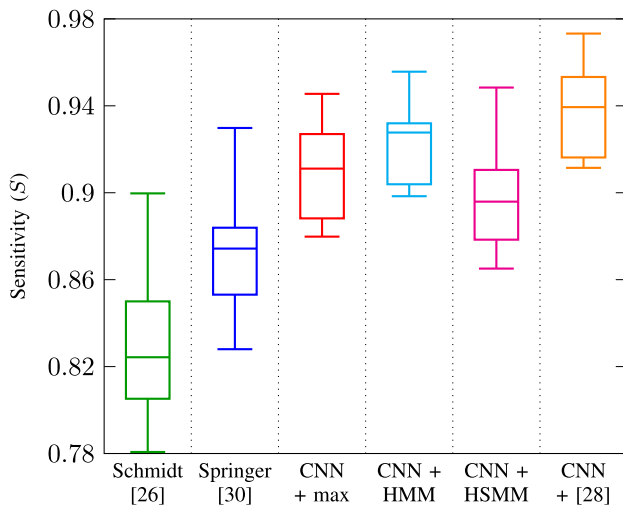


Fig. 7. Sensitivity (S) of the segmentation algorithm by Schmidt *et al.* [26], the algorithm by Springer *et al.* [30] and the proposed CNN-based approaches: CNN approach with sequential max temporal modeling (CNN + max), CNN approach combined with an HMM (CNN + HMM), CNN approach combined with an HSMM with sojourn time parameter estimation as in [30] (CNN + HSMM), CNN approach combined with an HSMM with sojourn time parameter estimation as in [28] (CNN + [28]).

setups, as proved by the p-values reported in Table II, which are obtained from paired sample t -tests run between the HSMM-based algorithm in [30] and each of the proposed CNN-based solutions.

D. Discussion

The numerical results shown in Section IV-C offered a comparison of the proposed approaches with other segmentation algorithms in the literature that use a temporal modulation model to provide information about the location and boundaries of fundamental heart sounds in PCG recordings. In this sense, the

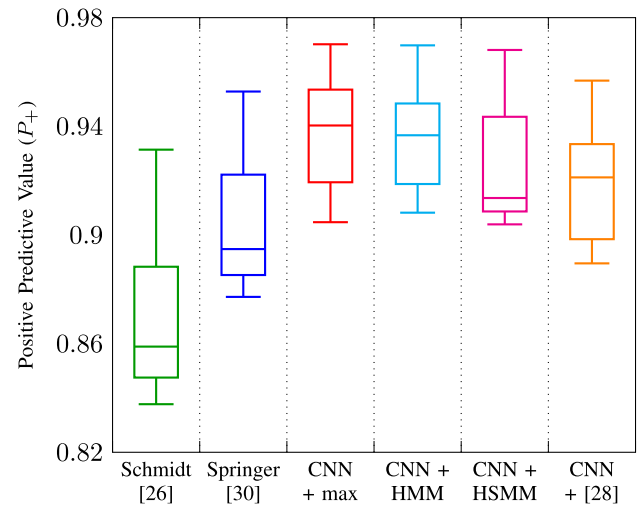


Fig. 8. Positive predictive value (P_+) of the segmentation algorithm by Schmidt *et al.* [26], the algorithm by Springer *et al.* [30] and the proposed CNN-based approaches: CNN approach with sequential max temporal modeling (CNN + max), CNN approach combined with an HMM (CNN + HMM), CNN approach combined with an HSMM with sojourn time parameter estimation as in [30] (CNN + HSMM), CNN approach combined with an HSMM with sojourn time parameter estimation as in [28] (CNN + [28]).

segmentation task considered in this work is in practice equivalent to that of classifying each sample from a PCG recording as belonging to an S1 sound, a systole interval, an S2 sound, or a diastole interval. Note that such task is significantly different from that of implementing a binary classifier for S1 and S2 sounds, which provides labels for features extracted from segments of the PCG that were previously selected as potential S1 and S2 sounds candidates. Then, when considering binary S1/S2 classifiers, such segments are usually obtained by applying peak-picking algorithms to the PCG signal (see, e.g., [20]). Therefore, the overall performance in segmenting heart sounds using this kind of approaches is affected by both the performance of the peak-picking algorithm adopted and by the specific binary classifier used to label candidate segments as S1 or S2 sounds. In addition, the performance of segmentation algorithms based on temporal modulation and that of binary heart sound classifiers is assessed with different methods, since temporal modulation algorithms are evaluated in terms of their capabilities in precisely locating the position of fundamental heart sounds, and not only in assigning the correct label to pre-selected segments. For these reasons, the numerical comparison included in this work has considered segmentation approaches based on temporal modulation, which attack the same problem as the proposed segmentation methods and that can be assessed using the same performance metrics.

The results reported in Section IV-C hint to the fact that the proposed CNN-based algorithms can capture more effectively the inter- and intra-sound state variability than the emission models used in [26] and [30]. Namely, the logistic regression-based emissions adopted in [30], jointly with a modified Viterbi algorithm that addresses boundary conditions, can guarantee better performance than the Gaussian distribution used to model emissions in [26]. On the other hand, CNNs are shown to model

TABLE I

EXPERIMENTAL RESULTS FOR DIFFERENT VALUES OF PATCH DIMENSION N AND STRIDE τ . THE PERFORMANCE METRICS CONSIDERED ARE SAMPLE ACCURACY (A), SENSITIVITY (S), AND POSITIVE PREDICTIVE VALUE (P_+). THE PERFORMANCE METRIC MEANS AND STANDARD DEVIATIONS COMPUTED FROM 10-FOLD CROSS-VALIDATION OVER THE HEART SOUND DATASET ARE REPORTED IN PERCENTAGE (%). BOLD VALUES CORRESPOND TO THE BEST VALUE ACHIEVED FOR A GIVEN METRIC IN A GIVEN EXPERIMENTAL SETUP

	$N = 64, \tau = 8$			$N = 128, \tau = 16$		
	A	S	P_+	A	S	P_+
Schmidt [26]	82.5 $\pm$ 2.8	83.3 $\pm$ 3.5	87.3 $\pm$ 3.1	84.5 $\pm$ 2.8	87.1 $\pm$ 3.7	89.7 $\pm$ 3.8
Springer [30]	86.0 $\pm$ 2.5	87.9 $\pm$ 3.2	90.8 $\pm$ 2.7	87.2 $\pm$ 2.1	89.8 $\pm$ 3.2	92.1 $\pm$ 2.8
CNN+max	91.5 $\pm$ 1.6	91.2 $\pm$ 2.3	94.1 $\pm$ 2.1	92.6 $\pm$ 1.6	92.7 $\pm$ 2.0	95.6 $\pm$ 2.0
CNN+HMM	91.8 $\pm$ 1.6	92.5 $\pm$ 2.0	93.8 $\pm$ 2.0	92.8 $\pm$ 1.4	93.7 $\pm$ 1.6	94.5 $\pm$ 1.9
CNN+HSMM	88.8 $\pm$ 1.7	90.1 $\pm$ 2.6	93.0 $\pm$ 2.4	89.7 $\pm$ 1.4	91.4 $\pm$ 1.9	93.6 $\pm$ 1.9
CNN + [28]	89.6 $\pm$ 1.5	93.9 $\pm$ 2.2	92.1 $\pm$ 2.2	90.6 $\pm$ 1.4	94.7 $\pm$ 1.8	93.7 $\pm$ 1.6
	$N = 256, \tau = 32$			$N = 512, \tau = 64$		
	A	S	P_+	A	S	P_+
Schmidt [26]	85.4 $\pm$ 3.4	89.4 $\pm$ 4.6	91.1 $\pm$ 3.8	87.4 $\pm$ 2.6	92.7 $\pm$ 3.1	93.3 $\pm$ 2.8
Springer [30]	88.1 $\pm$ 2.4	91.6 $\pm$ 3.2	93.2 $\pm$ 2.7	89.8 $\pm$ 1.2	94.3 $\pm$ 1.8	94.8 $\pm$ 1.8
CNN+max	93.0 $\pm$ 1.7	94.3 $\pm$ 1.9	95.4 $\pm$ 2.0	93.7 $\pm$ 1.0	95.2 $\pm$ 1.2	95.8 $\pm$ 1.4
CNN+HMM	93.2 $\pm$ 1.6	95.1 $\pm$ 1.8	95.3 $\pm$ 2.1	93.7 $\pm$ 1.0	95.7 $\pm$ 1.2	95.7 $\pm$ 1.4
CNN+HSMM	90.1 $\pm$ 2.0	93.0 $\pm$ 2.4	94.4 $\pm$ 1.8	91.5 $\pm$ 1.3	95.1 $\pm$ 1.0	95.4 $\pm$ 1.0
CNN + [28]	91.1 $\pm$ 1.7	95.3 $\pm$ 1.9	94.9 $\pm$ 1.8	91.7 $\pm$ 1.0	95.8 $\pm$ 1.0	95.7 $\pm$ 1.0

TABLE II

P-VALUES CORRESPONDING TO THE PAIRED t -TESTS APPLIED TO THE PERFORMANCE METRICS (A , S , AND P_+) ACHIEVED BY THE HSMM-BASED ALGORITHM IN [30] AND THE PROPOSED CNN-BASED SOLUTIONS. RESULTS CORRESPONDING TO DIFFERENT VALUES OF PATCH DIMENSION N AND STRIDE τ . BOLD VALUES CORRESPOND TO LACK OF STATISTICALLY SIGNIFICANT DIFFERENCE WITH RESPECT TO THE ALGORITHM IN [30], I.E., P-VALUES > 0.05

	$N = 64, \tau = 8$			$N = 128, \tau = 16$		
	A	S	P_+	A	S	P_+
CNN+max	7.6×10^{-6}	8.2×10^{-4}	3.6×10^{-4}	5.6×10^{-6}	6.8×10^{-3}	2.4×10^{-3}
CNN+HMM	5.6×10^{-6}	2.1×10^{-4}	1.1×10^{-3}	4.1×10^{-6}	1.2×10^{-3}	4.1×10^{-3}
CNN+HSMM	1.4×10^{-4}	2.2×10^{-4}	1.6×10^{-4}	1.7×10^{-5}	4.9×10^{-3}	5.4×10^{-3}
CNN + [28]	4.0×10^{-4}	6.5×10^{-5}	6.3×10^{-2}	2.5×10^{-4}	5.4×10^{-4}	6.3×10^{-2}
	$N = 256, \tau = 32$			$N = 512, \tau = 64$		
	A	S	P_+	A	S	P_+
CNN+max	3.9×10^{-6}	7.1×10^{-3}	4.2×10^{-3}	1.9×10^{-6}	6.9×10^{-2}	3.3×10^{-2}
CNN+HMM	3.7×10^{-6}	1.9×10^{-3}	5.5×10^{-3}	1.1×10^{-6}	4.6×10^{-3}	3.0×10^{-2}
CNN+HSMM	2.0×10^{-6}	1.2×10^{-2}	1.5×10^{-2}	1.2×10^{-3}	1.2×10^{-1}	2.3×10^{-1}
CNN + [28]	3.9×10^{-6}	1.3×10^{-3}	2.5×10^{-2}	6.1×10^{-4}	1.2×10^{-2}	6.5×10^{-2}

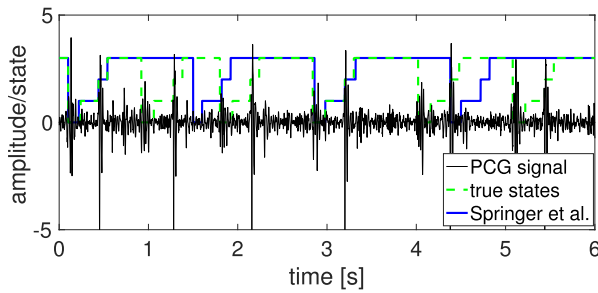
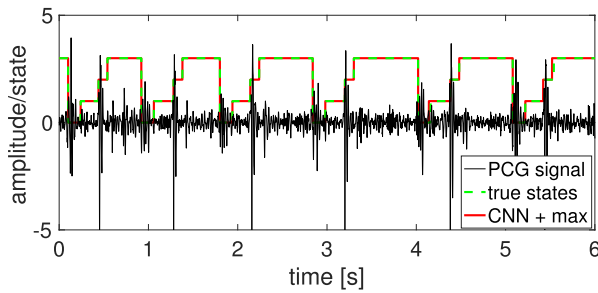
the complex behavior of heart sounds in a more refined manner. Moreover, the network architecture described in Section III-B provides a large receptive field which guarantees to incorporate effectively information coming from neighboring samples when assigning a given sample to the corresponding state. In this sense, CNNs are shown to implicitly model time evolution of the signal (actually of the patch) effectively. In fact, the proposed solutions allow for a more suitable modeling of sojourn time with respect to the segmentation algorithms in [26] and [30].

This behavior can be clearly observed in Fig. 9, which reports a segmentation example of a PCG from the PhysioNet dataset. It is possible to notice how the algorithm by Springer *et al.* returns an estimated state sequence with larger diastole duration than the true state sequence. This means that a poor initialization of the HSMM-based algorithm in [30], and in particular of the sojourn time statistics, can lead to significant errors, due to the lack of flexibility of this approach in modeling sojourn times.⁵

⁵In this case, the segmentation algorithm in [26] shows a behavior similar to that of the algorithm in [30] (not reported here for reasons of space), since both solutions are based on the adoption of an HSMM with sojourn time statistics that are derived from the test PCG using some heuristics.

On the other hand, the proposed CNN-solution in conjunction with the temporal modeling procedure described in Section III-C1 identifies almost perfectly the true state sequence of the heart sound signal reported in Fig. 9.

Numerical results also suggest that combining CNN outputs with an underlying HMM can improve the performance of the segmentation algorithm with respect to the temporal modeling solution proposed in Section III-C1, especially with respect to sample accuracy and sensitivity. On the other hand, a simple combination of the proposed CNN-based emission modeling with an underlying HSMM (with the sojourn time parameters heuristics used in [30]) seems to achieve worse performance than that guaranteed by the approach considering CNNs in conjunction with an HMM. This phenomenon might be considered perhaps as surprising, as the adoption of HSMMs (or, equivalently, duration-dependent HMMs) *in lieu* of HMMs for heart sound segmentation has been shown to boost segmentation performance considerably [25], [26]. However, it is possible to explain such results in terms of possible inaccuracies in the modeling of sojourn times when using the heuristics adopted by [26] and [30]. Also in this case, however, using CNNs to model emissions is shown to guarantee better results than the methods

(a) HSM-based algorithm by Springer *et al.* [30]

(b) Proposed CNN + sequential max segmentation algorithm

Fig. 9. Segmentation example: PCG signal (black lines), true states derived from the dataset annotations (green, dashed lines), states assigned by the HSM-based algorithm in [30] ((a), blue line), and states assigned by the proposed CNN-based segmentation algorithm with sequential max temporal modeling ((b), red line).

that models emissions either with a Gaussian distribution or via the logistic regression function. Moreover, the performance of the proposed CNN-HSM segmentation algorithm can be improved by refining the sojourn time parameter estimates via the approach presented in [28]. In fact, such solution is shown to achieve the highest sensitivity among all considered methods.

All the segmentation algorithms considered in the numerical results comparison were implemented on a workstation with an intel 6-core processor, 64 GB of RAM and mounting an Nvidia Titan Xp GPU. The average times required by the considered methods to segment a heart sound of duration 30 seconds were the following: 0.81 seconds for the method by Schmidt *et al.* [26], 1.01 seconds for the method by Springer *et al.* [30], 0.04 seconds for the CNN approach with sequential max temporal modeling, 0.23 seconds for the CNN approach combined with an HMM, 0.89 seconds for the CNN approach combined with an HSM, and 51.66 seconds for the CNN approach combined with in HSM with sojourn time parameter estimation in [28]. Note that this last method has a computational complexity significantly larger than the other methods due to the repeated use of EM iterations in the estimation of sojourn time parameters. It is also possible to note that the proposed CNN method in conjunction with the simple temporal modeling procedure described in Section III-C1 has the potential to offer segmentation in real time. This is guaranteed by the fact that different signal patches can be processed independently, without requiring the knowledge of the entire sequence. Moreover, all patches can be processed efficiently by leveraging current solutions for the computation of the outputs of convolutional neural networks, especially using GPU hardware. Note also that all operations

implemented in the proposed network (including convolutions) are 1-dimensional, thus they do not represent a particularly demanding computational burden. On the other hand, the solution involving the use of an HMM and the use of the Viterbi algorithm requires the observation of the entire sequence before starting the decoding procedure and has a computational complexity which is of the order $\mathcal{O}(4T)$, where T is the number of samples of the considered PCG signal. Similarly, the Viterbi algorithm for HSMs has a computational complexity of order $\mathcal{O}(20DT)$, where D represents the maximum sojourn time for all states [42].

Finally, it is important to underline also possible limitations related to the proposed approach. In particular, for what regards the experimental methodology, it is relevant to note that segmentation performance has not been assessed using independent datasets for training and testing, but 10-fold cross-validation has been used over the PhysioNet dataset, taking care about the fact that training and testing data were mutually exclusive. Moreover, although the proposed methods show a significant improvement with respect to other solutions in the literature when tested over all heart sounds in the PhysioNet dataset, such difference is somehow reduced when testing the segmentation approaches only on signals longer than 10.2 seconds, as shown in Tables I and II.

V. CONCLUSION

In this paper, the use of deep convolutional neural networks for heart sound segmentation was proposed. In particular, a convolutional network architecture, which is inspired by a network used for image segmentation, was combined with a sequential temporal modeling procedure in order to provide segmentation results compatible with the sequential nature of heart sounds. In addition, CNNs were also incorporated in segmentation algorithms using HMMs and HSMs as an efficient way to model emission distributions.

The proposed CNN-based solutions have been shown to guarantee improved performance in recovering the exact positions of fundamental heart sounds in a phonocardiogram with respect to the current state-of-the-art, when considering tests including also short heart sounds contained in the PhysioNet dataset.

Possible future extensions stemming from this work include the development of a general learning framework that allows to jointly train the CNN and the HMM (or HSM) at the same time.

ACKNOWLEDGMENT

The authors acknowledge the support of NVIDIA Corporation with the donation of the Titan Xp GPU used for this research.

REFERENCES

- [1] A. Pease, "If the heart could speak," *Pictures Future*, pp. 60–61, 2001.
- [2] S. Mangione, "Cardiac auscultatory skills of physicians-in-training: A comparison of three english-speaking countries," *Amer. J. Med.*, vol. 110, no. 3, pp. 210–216, 2001.
- [3] G. D. Clifford *et al.*, "Recent advances in heart sound analysis," *Physiol. Meas.*, vol. 38, pp. E10–E25, 2017.

- [4] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assisted Intervention*, 2015, pp. 234–241.
- [5] C. Liu *et al.*, "An open access database for the evaluation of heart sound algorithms," *Physiol. Meas.*, vol. 37, no. 12, pp. 2181–2213, 2016.
- [6] H. Liang, S. Lukkarinen, and I. Hartimo, "Heart sound segmentation algorithm based on heart sound envelopogram," in *Proc. Comput. Cardiol. Conf.*, 1997, pp. 105–108.
- [7] S. Ari, P. Kumar, and G. Saha, "A robust heart sound segmentation algorithm for commonly occurring heart valve diseases," *J. Med. Eng. Technol.*, vol. 32, no. 6, pp. 456–465, 2008.
- [8] A. Moukadem, A. Dieterlen, N. Hueber, and C. Brandt, "A robust heart sounds segmentation module based on S-transform," *Biomed. Signal Process. Control*, vol. 8, pp. 273–281, 2013.
- [9] S. Sun, Z. Jiang, H. Wang, and Y. Fang, "Automatic moment segmentation and peak detection analysis of heart sound pattern via short-time modified Hilbert transform," *Comput. Methods Programs Biomed.*, vol. 114, no. 3, pp. 219–230, 2014.
- [10] L. Huiying, L. Sakari, and H. Iiro, "A heart sound segmentation algorithm using wavelet decomposition and reconstruction," in *Proc. 19th Annu. Int. IEEE Eng. Med. Biol. Soc.*, Oct. 1997, vol. 4, pp. 1630–1633.
- [11] A. Castro, T. T. V. Vinhoza, S. S. Mattos, and M. T. Coimbra, "Heart sound segmentation of pediatric auscultations using wavelet analysis," in *Proc. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Jul. 2013, pp. 3909–3912.
- [12] H. Naseri and M. Homaeinezhad, "Detection and boundary identification of phonocardiogram sounds using an expert frequency-energy based metric," *Ann. Biomed. Eng.*, vol. 41, no. 2, pp. 279–292, 2013.
- [13] D. Kumar *et al.*, "Detection of S1 and S2 heart sounds by high frequency signatures," in *Proc. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2006, pp. 1410–1416.
- [14] J. Vepa, "Classification of heart murmurs using cepstral features and support vector machines," in *Proc. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2009, pp. 2539–2542.
- [15] C. N. Gupta, R. Palaniappan, S. Swaminathan, and S. M. Krishnan, "Neural network classification of homomorphic segmented heart sounds," *Appl. Soft Comput.*, vol. 7, no. 1, pp. 286–297, 2007.
- [16] T. Chen, K. Kuan, L. A. Celi, and G. D. Clifford, "Intelligent heart sound diagnostics on a cellphone using a hands-free kit," in *Proc. AAAI Spring Symp., Artif. Intell. Develop.*, 2009, pp. 26–31.
- [17] A. C. Stasis, E. Loukis, S. Pavlopoulos, and D. Koutsouris, "Using decision tree algorithms as a basis for a heart sound diagnosis decision support system," in *Proc. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2003, pp. 354–357.
- [18] J. E. Hebden and J. Torry, "Neural network and conventional classifiers to distinguish between first and second heart sounds," in *Proc. IEEE Colloq. Artif. Intell. Methods Biomed. Data Process.*, 1996, pp. 1–6.
- [19] A. A. Sepehri, A. Gharehbaghi, T. Dutoit, A. Kocharian, and A. Kiani, "A novel method for pediatric heart sound segmentation without using the ecg," *Comput. Methods Programs Biomed.*, vol. 99, no. 1, pp. 43–48, 2010.
- [20] T.-E. Chen *et al.*, "S1 and S2 heart sound recognition using deep neural networks," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 2, pp. 372–380, Feb. 2017.
- [21] H. Liu, H. Bai, L. He, and Z. Xu, "Stochastic sequential neural networks with structured inference," 2017, arXiv:1705.08695.
- [22] L. Gamero and R. Watrous, "Detection of the first and second heart sound using probabilistic models," in *Proc. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2003, pp. 2877–2880.
- [23] D. Gill, N. Gavrieli, and N. Intrator, "Detection and identification of heart sounds using homomorphic envelopogram and self-organizing probabilistic model," in *Proc. Comput. Cardiol. Conf.*, 2005, pp. 957–960.
- [24] Y.-J. Chung, "Classification of continuous heart sound signals using the ergodic hidden Markov model," in *Iberian Conf. Pattern Recog. Image Anal.*, Berlin, Heidelberg: Springer, 2007, pp. 563–570.
- [25] S. Schmidt, E. Toft, C. Holst-Hansen, C. Graff, and J. Struijk, "Segmentation of heart sound recordings from an electronic stethoscope by a duration dependent hidden-Markov model," in *Proc. Comput. Cardiol. Conf.*, 2008, pp. 345–348.
- [26] S. Schmidt, C. Holst-Hansen, C. Graff, E. Toft, and J. J. Struijk, "Segmentation of heart sound recordings by a duration-dependent hidden Markov model," *Physiol. Meas.*, vol. 31, no. 4, pp. 513–529, 2010.
- [27] J. Oliveira, T. Mantadelis, F. Renna, P. Gomes, and M. Coimbra, "On modifying the temporal modeling of HSMMs for pediatric heart sound segmentation," in *Proc. IEEE Int. Workshop Signal Process. Syst.*, Oct. 2017, pp. 1–6.
- [28] J. Oliveira, F. Renna, and M. T. Coimbra, "Adaptive sojourn time HSMM for heart sound segmentation," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 2, Mar. 2019.
- [29] D. B. Springer, L. Tarassenko, and G. D. Clifford, "Support vector machine hidden semi-Markov model-based heart sound segmentation," in *Proc. Comput. Cardiol. Conf.*, Sep. 2014, pp. 625–628.
- [30] D. B. Springer, L. Tarassenko, and G. D. Clifford, "Logistic regression-HSMM-based heart sound segmentation," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 4, pp. 822–832, Apr. 2016.
- [31] C. Liu, D. Springer, and G. D. Clifford, "Performance of an open-source heart sound segmentation algorithm on eight independent databases," *Physiol. Meas.*, vol. 38, no. 8, pp. 1730–1745, 2017.
- [32] I. Grzegorzczak *et al.*, "PCG classification using a neural network approach," in *Proc. Comput. Cardiol. Conf.*, Sep. 2016, pp. 1129–1132.
- [33] E. Kay and A. Agarwal, "Dropconnected neural network trained with diverse features for classifying heart sounds," in *Proc. Comput. Cardiology Conf.*, Sep. 2016, pp. 617–620.
- [34] T. Nilanon, J. Yao, J. Hao, S. Purushotham, and Y. Liu, "Normal/abnormal heart sound recordings classification using convolutional neural network," in *Proc. Comput. Cardiol. Conf.*, 2016, pp. 585–588.
- [35] J. Rubin, R. Abreu, A. Ganguli, S. Nelaturi, I. Matei, and K. Sricharan, "Classifying heart sound recordings using deep convolutional neural networks and mel-frequency cepstral coefficients," in *Proc. Comput. Cardiol. Conf.*, 2016, pp. 813–816.
- [36] H. Ryu, J. Park, and H. Shin, "Classification of heart sound recordings using convolution neural network," in *Proc. Comput. Cardiol. Conf.*, 2016, pp. 1153–1156.
- [37] D. M. Nogueira, C. A. Ferreira, and A. M. Jorge, "Classifying heart sounds using images of MFCC and temporal features," in *Proc. Portuguese Conf. Artif. Intell.*, Springer, 2017, pp. 186–203.
- [38] O. Deperlioglu, "Classification of phonocardiograms with convolutional neural networks," *Broad Res. Artif. Intell. Neurosci.*, vol. 9, no. 2, pp. 22–33, 2018.
- [39] V. G. Sujadevi, K. P. Soman, R. Vinayakumar, and A. U. Prem Sankar, "Anomaly detection in phonocardiogram employing deep learning," in *Computational Intelligence in Data Mining. Advances in Intelligent Systems and Computing*, vol. 711, H. Behera, J. Nayak, B. Naik, and A. Abraham, Eds. Singapore: Springer, 2019, pp. 525–534.
- [40] I. Goodfellow, Y. Bengio, A. Courville, and Y. Bengio, *Deep Learning*, vol. 1. Cambridge, MA, USA: MIT Press, 2016.
- [41] C. Bishop, *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Secaucus, NJ, USA: Springer-Verlag, 2006.
- [42] S.-Z. Yu, "Hidden semi-Markov models," *Artificial Intell.*, vol. 174, no. 2, pp. 215–243, 2010.
- [43] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, arXiv:1412.6980.
- [44] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. Int. Conf. Artif. Intell. Statist.*, 2010, pp. 249–256.
- [45] F. Chollet *et al.*, "Keras," 2015. [Online]. Available: <https://keras.io>
- [46] M. Abadi *et al.*, "TensorFlow: Large-scale machine learning on heterogeneous systems," 2015. [Online]. Available: <http://tensorflow.org/>